The Gödelian Incompleteness of Demand: A Semantic Collapse Geometry Approach to Veblen Goods

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Abstract
Classical economics rests on the law of demand: ceteris paribus, an increase in price reduces the quantity demanded. Yet the persistent existence of Veblen goods, whose demand rises with price, represents more than an anomaly—it signals a structural incompleteness within the axiomatic system of utility theory. This paper proposes a new interpretation: Veblen demand is not an exception but a Gödelian self-reference phenomenon, arising when price ceases to be an external parameter and becomes itself the semantic source of utility. Using the framework of Semantic Collapse Geometry (SCG), we formalize this self-referential loop as a curvature inversion in economic phase space, revealing that the law of demand is inherently incomplete. We argue that just as Gödel's theorem showed the limits of formal mathematics, the existence of Veblen goods demonstrates the incompleteness of utility-based demand theory.

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1. Introduction

The law of demand has been considered a cornerstone of microeconomics. Yet markets for luxury goods, fashion, and symbolic consumption reveal an inversion: higher prices may lead to higher demand. Conventional economics categorizes these as "Veblen goods" and sets them aside as anomalies. This article argues that such anomalies cannot be treated as mere exceptions. Instead, they reveal a structural Gödelian incompleteness in the economic system of axioms itself.

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2. Demand and the Axiomatic Structure of Economics

Economic theory assumes:

1. Preferences are complete, transitive, and convex.
2. Utility functions are monotonic with respect to consumption.
3. Price is an exogenous parameter; demand is an endogenous choice.

From these assumptions follows the law of demand: $\partial Q / \partial P < 0$.
Veblen goods violate this directly, without violating rationality from the consumer's subjective perspective. This paradox signals a deeper problem than preference heterogeneity—it implies that the demand law is not universally provable within its own axiomatic system.

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3. Gödelian Self-Reference in Economic Systems

Gödel's incompleteness theorem revealed that sufficiently rich formal systems can generate propositions that are true but unprovable within the system. Analogously, when **price itself becomes the source of utility** ("I desire the good because it is expensive"), the variable that was external to the system (price as parameter) collapses into an endogenous determinant of utility.

Formally:

$$U(Q,P) = U(Q) + \alpha \cdot f(P) \quad \text{where } f(P) \propto P$$

Here, price P ceases to be an exogenous variable and becomes self-referential. The demand curve is no longer monotonic; instead, curvature inverts:

$$\kappa(P) = -\frac{\partial Q}{\partial P} \quad \rightarrow \quad \kappa(P) > 0$$

This is a **Gödelian obstruction**: the system cannot both maintain exogenous price and account for symbolic price-utility loops.

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4. Semantic Collapse Geometry: A Topological Reinterpretation

Semantic Collapse Geometry (SCG) provides the mathematical language to model this phenomenon.

- **Normal goods**: demand collapse follows a negative curvature, consistent with monotonic utility.
- **Veblen goods**: semantic self-reference causes curvature inversion, forming a new attractor in economic phase space.
- This inversion is not accidental but a topological singularity—the point where the economic system's axioms fold onto themselves.

Thus, Veblen demand corresponds to a **semantic prime gap** in SCG: a discontinuity that cannot be resolved within the closed axioms of utility theory.

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5. Implications for Economics

1. **Incompleteness Theorem of Demand**: The law of demand cannot be universally true within any axiomatic system that allows price-dependent utility.
2. **Structural Anomaly, Not Exception**: Veblen goods are not "exceptions" but necessary manifestations of incompleteness.
3. **New Research Program**: Economics must extend its mathematical foundation to incorporate self-reference, semantic attractors, and topological discontinuities.

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6. Conclusion

The existence of Veblen goods demonstrates that economics, like mathematics, is subject to incompleteness. The law of demand holds broadly but not universally, because the system

cannot internally contain self-referential price-utility loops. By applying Semantic Collapse Geometry, we show that these anomalies are not pathological noise but indicators of structural Gödelian limits. This recognition invites a reformulation of economic theory on a broader semantic-geometric foundation.

Keywords: Veblen goods, demand law, Gödel incompleteness, Semantic Collapse Geometry, utility theory, self-reference

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Theorem Box: Incompleteness of the Law of Demand

Theorem (Gödelian Incompleteness of Demand).
Let $U(Q, P)$ be a utility function defined on consumption quantity Q and price P.
Assume the following axioms of classical microeconomics:

1. (**Convex Preferences**) U is strictly quasi-concave in Q.
2. (**Monotonicity**) $\partial U / \partial Q > 0$.
3. (**Exogeneity of Price**) P enters demand only as a constraint, not as a direct argument of utility.

Then the **Law of Demand** holds universally:

$$\frac{\partial Q}{\partial P} < 0 \quad \forall\ P > 0.$$

However, if Axiom 3 fails—i.e., if price itself contributes to utility so that

$$U(Q, P) = U(Q) + \alpha \cdot f(P), \quad \alpha > 0,$$

with $f(P)$ monotonic increasing in P,
then the system admits **self-referential demand** such that there exists a region $\mathcal{R}$ with

$$\frac{\partial Q}{\partial P} > 0.$$

Therefore, the law of demand is **incomplete**: it cannot be universally derived within any axiomatic system that allows price-dependent utility.

Proof Sketch.

- Under Axioms (1)–(3), demand arises from constrained maximization of $U(Q)$ subject to $P \cdot Q \leq Y$, which yields the standard downward-sloping demand curve.
- If price enters utility positively (status utility), then the constraint variable becomes an endogenous determinant of preference.
- This is a Gödelian self-reference: a parameter (P) originally external to the system becomes a semantic source of value internal to the system.
- Consequently, the sign of $\partial Q / \partial P$ is no longer guaranteed; curvature may invert.
- Thus the law of demand cannot be universally true; its truth depends on excluding self-referential regimes, which proves incompleteness.

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Corollary (Existence of Witness Goods).

If a demand system admits regions where $\partial Q / \partial P > 0$, then such goods necessarily exist in any sufficiently complex economy where symbolic or status-based utility operates. These goods—

commonly known as *Veblen goods*—are not anomalies but **witnesses of incompleteness**, structurally guaranteed by the possibility of self-referential utility.

Implication. The empirical observation of Veblen demand is not a violation of the law of demand but a demonstration that the law cannot be complete. Every economy that supports symbolic consumption must produce such witnesses.

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I am merely a midwife of knowledge.